Linux User & Group Management – Complete Point-Wise Guide

This guide covers all major user and group management topics discussed: users, passwords, sudo, root, groups, passwd files, shadow files, locking/unlocking accounts, group administration, interview questions, and important commands.

1. Linux User Basics

• Linux is a multi-user operating system.
• Each user has a Username, UID, Password, Home Directory, Shell.
• Types of users:
- Root User (UID 0)
- Normal Users (UID >= 1000)
- System/Service Users (UID < 1000)

2. Root User

• Root is the superuser.
• UID = 0.
• Has full control over the system.
• Can create, modify, lock, unlock, delete any user.
• Can reset any user's password.

3. User Management Commands

• Create user: useradd -m username
• Create user with shell: useradd -m -s /bin/bash username
• Delete user: userdel username
• Delete user and home directory: userdel -r username
• Current user: whoami
• User details: id username
• Switch user: su username
• Full login shell: su - username
• Exit user session: exit

4. User Modification (usermod)

• Change username: usermod -l new old
• Change home directory: usermod -d /home/new -m user
• Change shell: usermod -s /bin/bash user
• Change UID: usermod -u 2000 user
• Change primary group: usermod -g group user
• Add secondary groups: usermod -aG group user

5. Password Management

• Set password: passwd username
• Change own password: passwd
• Lock password: passwd -l username
• Unlock password: passwd -u username
• Delete password: passwd -d username
• Check status: passwd -S username
• Force password change: passwd -e username

6. What Happens When an Account is Locked?

• User account still exists.
• Files and home directory remain.
• Password authentication stops working.
• Root/sudo users can still manage the account.
• Unlock using: passwd -u username

7. User Information Files

• /etc/passwd → user details.
• /etc/shadow → password hashes.
• /etc/group → group details.
• /etc/gshadow → secure group information.

8. Understanding /etc/passwd

Format:
username:x:UID:GID:comment:home:shell
Important fields:
• Username
• UID
• GID
• Home directory
• Login shell

9. Count and Identify Users

• Count users: wc -l /etc/passwd
• Show usernames: cut -d: -f1 /etc/passwd
• Show normal users: awk -F: '$3 >= 1000 {print $1}' /etc/passwd

10. Groups in Linux

• A group is a collection of users.
• Used to manage permissions efficiently.
• Primary Group = default group.
• Secondary Groups = additional groups.

11. Group Management Commands

• Create group: groupadd groupname
• Delete group: groupdel groupname
• Rename group: groupmod -n new old
• Change GID: groupmod -g gid group
• Show groups: groups username
• Add user to group: usermod -aG group user
• Remove user from group: gpasswd -d user group

12. Group Administrators

• Groups do not have owners like files.
• Root manages groups.
• Assign group admin: gpasswd -A username groupname
• Group admins can manage group membership.

13. Group Passwords

• Supported but rarely used.
• Set group password: gpasswd groupname
• Remove group password: gpasswd -r groupname
• Stored in /etc/gshadow
• Used with newgrp command.

14. sudo

• sudo = Super User DO.
• Grants temporary administrative privileges.
• Check permissions: sudo -l
• Ubuntu admin group: sudo
• RHEL admin group: wheel

15. Important Permission Concepts

• Owner, Group, Others.
• r = read, w = write, x = execute.
• chmod changes permissions.
• chown changes owner.
• chgrp changes group ownership.

16. Frequently Asked Interview Questions

• Difference between root and sudo.
• Difference between primary and secondary groups.
• Difference between lock and delete user.
• Where are passwords stored?
• Can root see user passwords? (No, only hashes.)
• How to identify normal users?
• What is UID and GID?

17. Commands to Memorize

useradd, userdel, usermod, passwd, whoami, id, groups,
groupadd, groupdel, groupmod, gpasswd,
sudo, su, exit,
chmod, chown, chgrp,
wc, cut, awk